

Research on Experimental Design and Statistical Inference of Agricultural Sensor Data

I. Introduction and Project Background

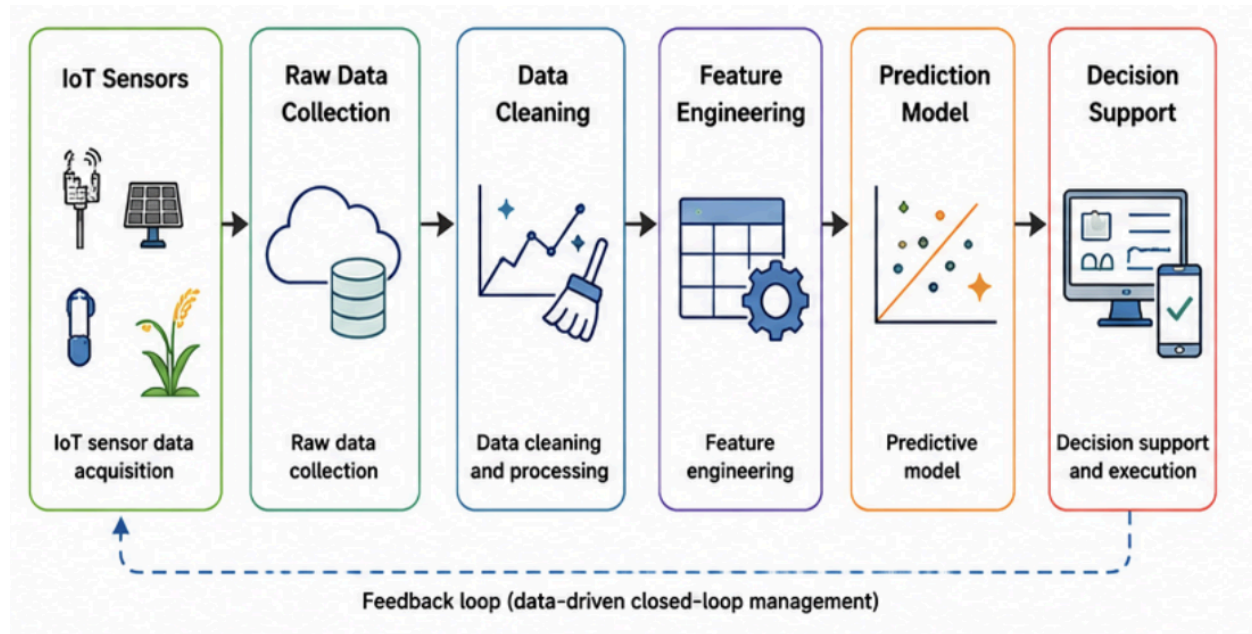
With the continuous advancement of agricultural digital transformation, the Internet of Things (IoT), artificial intelligence, and machine learning technologies have gradually become vital research directions in smart agriculture. Research indicates that high-frequency environmental monitoring data can effectively enhance crop yield prediction, quality assessment, and agricultural resource management capabilities. However, most existing studies focus solely on improving prediction accuracy, while research on how to further translate prediction results into business decisions and resource optimization remains relatively limited (Taha et al., 2025). In agricultural supply chain management practices, decision-makers are more concerned with specific management problems, such as when to irrigate, when to fertilize, when to issue risk warnings, and how to maximize returns. As emphasized by the latest digital agriculture research, the future developmental direction of smart agriculture should progressively shift from being prediction-oriented to decision-oriented, achieving a deep integration of data-driven management and resource optimization (Botero-Valencia et al., 2025; Miller et al., 2025; Shamshiri et al., 2024).

This project intends to construct a high-frequency agricultural data collection and analysis platform on a rice experimental field of approximately 10 mu. This study not only aims to establish a yield and quality prediction system through multimodal environmental data but, more importantly, seeks to build a comprehensive prediction-decision closed-loop business analytics framework. By combining statistical machine learning algorithms with asymmetric cost control theory (Thomas & Crook, 2024) and intertemporal stochastic control methods (Deng et al., 2025), this research explores digital agriculture operational modes that reduce revenue volatility in the agricultural supply chain, optimize resource allocation, and enhance asset value and quality stability.

The main contributions of this study are reflected in the following three aspects:

1. It constructs a decision-aware analytical framework oriented toward agricultural supply chain asset management, systematically integrating IoT data collection (Wang et al., 2025), machine learning prediction, and operational decision optimization.
2. Under small-sample agricultural experimental conditions (Gross et al., 2024), it combines high-frequency time-series monitoring data to establish a panel data augmentation analysis system and utilizes robust statistical inference methods (Kaufman & Goutte, 2026) to improve the reliability of the research findings.

3. It extends traditional agricultural prediction models into a decision support system geared toward resource allocation and revenue optimization, providing new research ideas and practical pathways for digital agriculture, smart agriculture, and agricultural supply chain risk management (Silva & Santos, 2025).



II. Goal Layer Design: Data Value Chain and Decision Model Positioning

2.1 Demand-Driven Analysis: Why Decision-Makers Use Data

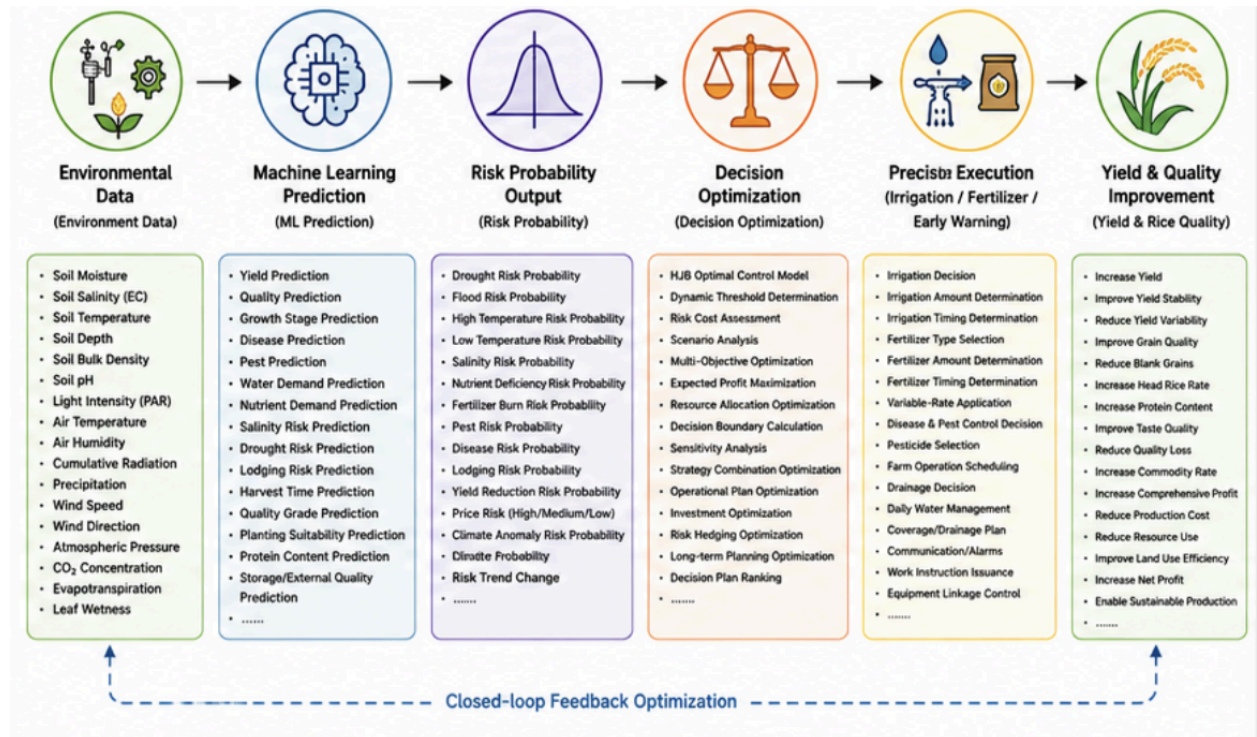
In modern food supply chain management, agricultural managers, supply chain risk management personnel, and commodity buyers introduce IoT high-frequency environmental data not merely to obtain yield prediction results, but to utilize data to bridge the chasm between prediction and decision-making, thereby achieving proactive risk prevention and control. First, continuous environmental monitoring enables quality optimization. Rice quality indicators (e.g., head rice recovery rate, chalkiness, protein content) exhibit high sensitivity to continuous environmental changes. Dynamic environmental perception can help management personnel mitigate external environmental shocks, allowing products to meet high-quality commercial standards more stably. Second, by quantifying the economic returns brought by different management measures under asymmetric cost constraints, the framework provides automated decision support, ensuring that the cost of proactive intervention remains far lower than the major economic losses that could result from inaction (Thomas & Crook, 2024), thereby maximizing overall economic benefits.

2.2 Business Intelligence Analysis: How Decision-Makers Process Data

To truly transform underlying sensor data into management decision bases, this study establishes a three-layer progressive analysis system.

- The first layer is time-series panel feature engineering. By applying sliding time windows to high-frequency environmental data for aggregation processing, critical variables such as the cumulative water stress index and microclimate fluctuation index are extracted, transforming continuous time series into structured analysis data (Deng et al., 2025).
- The second layer is predictive analysis and calibration evaluation. Logistic regression, random forest, and gradient boosting models are employed for prediction, and a comprehensive evaluation of model predictive performance and probability calibration capability is performed (Zhang & Johnstone, 2024) to ensure that prediction results can serve as an important basis for risk management.
- The third layer is the decision-aware optimization layer. Combined with dynamic control theory, prediction probabilities are mapped directly to optimal decision boundaries, realizing the automatic optimization of irrigation, water-fertilizer regulation, and risk management strategies.

In terms of technical implementation, data processing and analysis are primarily completed based on the Python data science toolchain, including Pandas for data processing, Scikit-Learn for machine learning modeling, and Power BI as a visual analysis platform. Data storage utilizes the InfluxDB time-series database and Snowflake structured data warehouse (Wang et al., 2025). Recent studies demonstrate that the deep integration of machine learning models and agricultural decision support systems can significantly improve agricultural management efficiency and enhance risk response capabilities under complex environmental conditions (Silva & Santos, 2025). Particularly in high-frequency environmental monitoring scenarios, a closed-loop analysis framework combining prediction models and decision optimization mechanisms is becoming an essential development trend in smart agriculture research (Botero-Valencia et al., 2025; Miller et al., 2025).



2.3 Data Asset Specifications: What Kind of Data is Needed

The platform consistently adopts a structured, tidy data format for storage, utilizing a time-series database at the backend to accomplish real-time parsing and data management. To guarantee subsequent machine learning modeling, statistical analysis, and visual displays, all data are uniformly stored in a structured data format. Original sensor data are uploaded in real time through IoT gateways, and stored in the time-series database and structured database after data cleaning and standardization (Wang et al., 2025), ensuring a uniform correspondence between data features and crop growth processes. The platform supports CSV, Parquet, and API interface outputs to fulfill subsequent modeling and business analysis demands.

Table 1: Data Storage Structure Design

Field Name	Data Type	Description
Timestamp	Datetime	Data collection time
Block ID	Integer	Plot number
Sensor ID	String	Device number
Metric	String	Monitoring metric name
Value	Float	Metric measurement value

Quality_Flag	Binary	Data quality flag
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Table 2: Environmental Variables and Business Meanings

Variable Category	Indicator	Management Significance
Soil Environment	Soil Moisture	Irrigation decision
Soil Environment	Soil Temperature	Root activity
Soil Environment	Soil EC	Salinity monitoring
Soil Environment	Soil pH	Nutrient availability
Meteorological Environment	Air Temperature	Growth rate
Meteorological Environment	Air Humidity	Pest and disease risk
Meteorological Environment	Light Intensity	Photosynthesis
Meteorological Environment	Precipitation	Water replenishment

Table 3: Research Response Variable Definitions

Response Variable	Unit	Description
Yield	kg/mu	Final yield
Premium Rice Ratio / Quality Grade / Head Rice Recovery	% / Grade / %	Premium commercial rice ratio / Quality rating / Head rice recovery rate
Protein Content	%	Protein content

Table 4: Rice Growth Stage Division

Stage	Time Range
Seedling Stage	Day 1-30
Tillering Stage	Day 31-50
Jointing and Booting Stage	Day 51-70
Heading and Grain-filling Stage	Day 71-105

Table 5: Decision Requirements and Data Value Mapping Table

Decision Problem	Data Used	Analysis Method	Decision Output
Whether to irrigate	Soil moisture, Precipitation	Threshold analysis	Irrigation initiation recommendation
Whether to fertilize	PH, EC	Regression analysis	Fertilization strategy
Whether to alert	Air temperature, Humidity	Risk prediction model	Risk notification
Quality enhancement	Environmental variable set	Machine learning model	Quality optimization recommendation
Revenue maximization	All variables	Decision-Aware optimization	Optimal management scheme

III. Execution Layer Design: Field IoT Engineering and Data Quality Control

To ensure that data collection can satisfy the demands of agricultural production management and decision optimization, this study constructs an indicator system by combining agricultural production mechanisms with business analysis requirements. Soil moisture is used to support irrigation decisions and drought risk identification; soil temperature reflects root activity and nutrient absorption capacity; soil electrical conductivity (EC) monitors soil salinity fluctuations; soil pH assesses nutrient availability; air temperature reflects crop growth velocity; air humidity monitors pest and disease occurrence risks; light intensity and cumulative radiation reflect photosynthesis levels; and precipitation assists in formulating irrigation management strategies. Together, these indicators constitute the core environmental variable system affecting rice yield and quality formation. High-frequency IoT monitoring can effectively capture the dynamic shocks caused by these microclimate fluctuations on crop growth (Han et al., 2025; Shamshiri et al., 2024).

3.1 Data Pipeline and Data Warehouse Architecture

In terms of technical implementation, this study constructs a stable transmission link based on industrial-grade environmental monitoring equipment. All sensor data, aggregated via wireless gateways, are transmitted back in real time to the cloud time-series database InfluxDB and the Snowflake structured data warehouse through the standard lightweight publish/subscribe Message Queuing Telemetry Transport (MQTT) communication protocol (Wang et al., 2025). Vertically, by embedding dual-layer probes at 10 cm and 20 cm, the system achieves dynamic monitoring of soil electrical conductivity (EC), thereby providing underlying data support for closed-loop control (Martínez-Gomez & Zhao, 2026).

3.2 Sampling Frequency Design Rationale

This study adopts a data sampling frequency of once every 30 minutes. The design of sampling frequency directly determines the trade-off between microclimate fidelity and node lifespan (Sambo & Qadir, 2025). If the sampling frequency is too low, critical environmental changes such as sudden high temperatures or heavy rainstorms cannot be captured in time, affecting the efficacy of risk warnings. If the sampling frequency is too high, because environmental variables change minimally over short durations, a large volume of highly correlated data will be generated, which not only increases system burdens but also raises equipment power consumption. Synthesizing information acquisition efficiency, equipment lifespan, and operational costs, a 30-minute sampling interval can effectively balance information loss and noise contamination in the discretization analysis of continuous time series (Deng et al., 2025), realizing an optimal cost-benefit equilibrium. For minor amounts of missing data that may occur within short intervals, the system will utilize cubic spline interpolation for automated imputation.

3.3 Spatial Randomized Block Design and Equipment Deployment

For a rice experimental field of approximately 10 mu in Panjin City, Liaoning Province, this study employs a spatial randomized block design to partition it into 12 independent spatial blocks. The size of a single spatial grid is approximately 23.5 m×23.5 m (about 0.83 mu). This randomized block design can effectively overcome spatial interference and violations of the Stable Unit Treatment Value Assumption (SUTVA, i.e., randomness of individual treatment effects) in field-level causal inference (Gross et al., 2024). Regarding the selection of specific deployment points, the study determines the optimal spatial grid scale for wireless sensor placement through anisotropic variogram analysis (Han et al., 2025).

In the experiment, a set of comprehensive environmental monitoring equipment is deployed at the geometric center of each block. Each set of equipment vertically embeds dual-layer probes at 10 cm and 20 cm underground to continuously monitor moisture, temperature, and salinity variations in the vertical soil direction. Through spatial randomized block design and standardized deployment, the system can effectively reduce statistical bias caused by spatial autocorrelation, substantially improving the reliability of subsequent machine learning model inferences and business intelligence analysis results.

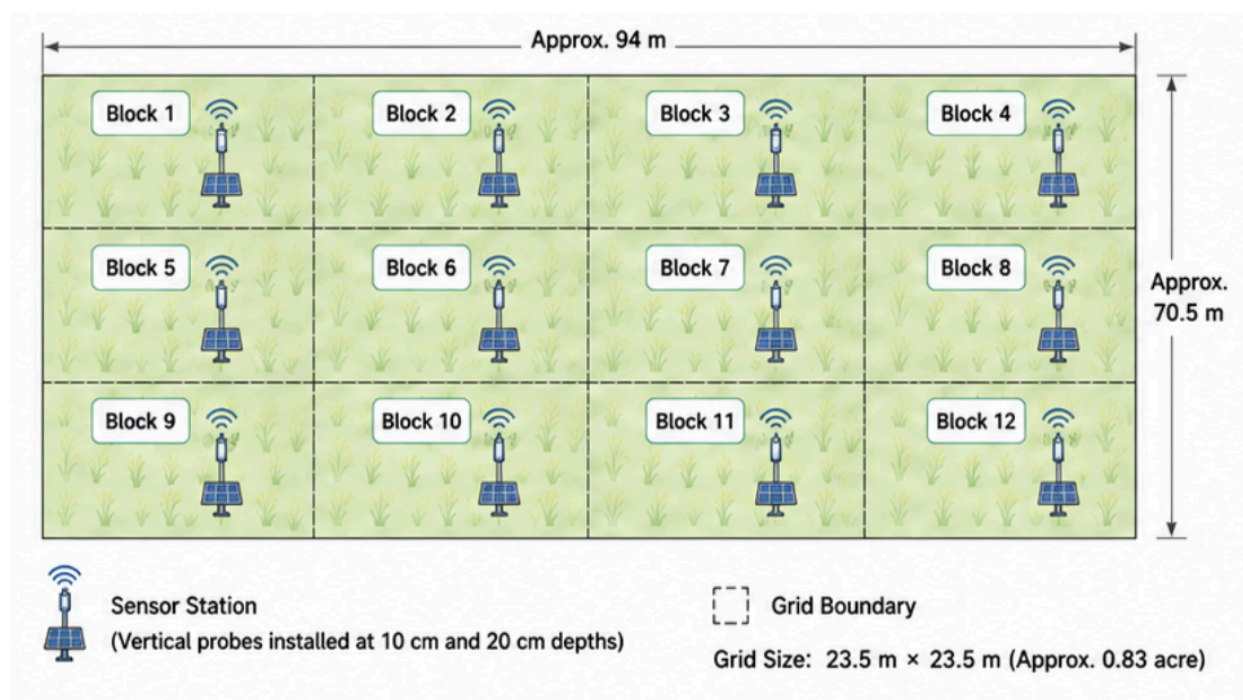


Table 6: Equipment Configuration and Deployment Scheme

Equipment Type	Monitoring Metric & Deployment Quantity	Deployment Location	Supplier
Soil Sensor	Temperature, Humidity, EC, PH / 12 sets	1 set per block	Topcloud-Y農 (Topu Yunong)
Weather Station	Temperature, Humidity, Precipitation / 3 sets	Field edges	Juying Electronics
LoRa Gateway	Data transmission / 3 sets	Center area	METER
Cloud Platform	InfluxDB / Snowflake / 1 set	Cloud	Cloud Service Provider

3.4 Equipment Procurement and Third-Party Maintenance System

The project utilizes industrial-grade agricultural environmental monitoring equipment uniformly configured with soil temperature, humidity, electrical conductivity, and pH integrated sensors, complemented by wireless communication gateways. Meanwhile, a long-term third-party operation and maintenance (O&M) system is established, which is regularly carried out by a cooperative agricultural IoT technical service team. The project team is responsible for data analysis and model development, while the third-party technical service team handles on-site equipment maintenance and system inspections. This ensures stable and reliable data collection throughout the entire life cycle (Sambo & Qadir, 2025).

Equipment procurement sources include domestic agricultural IoT equipment suppliers (Topu Yunong, Juying Electronics, etc.) as well as international agricultural monitoring equipment manufacturers (METER Group, Campbell Scientific, etc.). All devices support standard communication protocols, including LoRaWAN, MQTT, and REST API interfaces, and can directly ingest data into the InfluxDB time-series database and Snowflake data warehouse (Wang et al., 2025). This realizes continuous monitoring and closed-loop control of dual-layer soil microclimates (Martínez-Gomez & Zhao, 2026), ultimately achieving unified data management and analysis.

IV. Model Construction and Decision Optimization Framework

After completing agricultural IoT data collection, storage, and quality control, this study further constructs a decision-aware analytical framework oriented toward agricultural supply chain management, achieving full-process data-driven analysis from environmental monitoring to management decision optimization.

First, preprocessing is performed on high-frequency environmental variables such as soil moisture, soil temperature, soil EC, soil PH, air temperature, air humidity, light intensity, and precipitation. Combined with sliding time windows, feature variables such as cumulative precipitation, cumulative effective temperature, water stress index, and microclimate fluctuation index are calculated, thereby constructing a structured analysis dataset reflecting crop growth status.

At the predictive model layer, the study uses machine learning methods such as logistic regression, random forest, and gradient boosting models to predict rice yield, quality grades, and potential risk statuses. Cross-validation and probability calibration techniques are used during model training to comprehensively evaluate the accuracy, stability, and interpretability of prediction results (Zhang & Johnstone, 2024), ensuring that prediction results truly reflect risk variation trends in agricultural production.

Distinct from traditional agricultural research that only focuses on prediction accuracy, this study further introduces decision-aware analytical concepts, mapping model output results directly to actual management decision scenarios. When prediction results indicate that crops face a water stress risk, the system automatically generates irrigation recommendations; when soil nutrient levels fall below set thresholds, the system generates fertilization optimization schemes; when environmental conditions trigger pest and disease risk warning rules, the system issues timely warning information to management personnel. By tightly coupling prediction results with management measures, dynamic optimization of agricultural resource allocation is achieved. Ultimately, this study builds a closed-loop analytical mechanism of environmental perception → risk prediction → decision optimization → effect feedback. This empowers agricultural managers not only to understand problems that might occur in the future but also to adopt optimal

intervention measures based on prediction results. Consequently, it reduces production risks, improves rice quality stability and overall economic returns of the agricultural supply chain, and realizes a paradigm shift from prediction-oriented to decision-oriented data analysis modes.

V. Statistical Inference under Small Sample Conditions

5.1 Non-parametric Inference under Small Sample Panel Data Augmentation

Because the experimental platform is constrained by land resources and management costs, only 12 independent spatial blocks are set up, posing a typical small-sample agricultural experimental scenario. Under these conditions, traditional large-sample statistical inference methods that rely on asymptotic properties may yield substantial biases. To improve the robustness of inference outcomes, this study fully exploits the characteristic of IoT sensors capturing data automatically every 30 minutes. Throughout the rice growth cycle, environmental variables, management variables, and crop growth indicators are continuously accumulated, thereby constructing a high-frequency panel dataset with both temporal and spatial dimensions. Although the final yield and quality results originate from only 12 spatial samples, the massive volume of time-series observations effectively augments the information capacity for model training and parameter estimation.

In the model evaluation phase, this study utilizes a non-parametric paired bootstrap method for robust statistical inference. Specifically, by performing repetitive sampling with replacement across the 12 spatial blocks, the exact same set of samples is maintained for different models and decision strategies during each resampling process, thereby ensuring consistency and fairness in model comparison. This method effectively diminishes the impact of spatial random errors on experimental results, and further constructs bias-corrected confidence intervals to evaluate whether different decision strategies possess statistical significance in terms of yield enhancement, quality improvement, and economic return optimization. Compared to traditional single-run experimental result analyses, this approach can more robustly identify the actual value brought forth by data-driven decisions.

5.2 Expected Outcomes and Research Value

This study expects to construct a comprehensive data-driven decision system covering the entire process of agricultural production, including an agricultural IoT monitoring platform, an agricultural structured data management platform, a rice yield prediction model, a quality prediction model, an agricultural intelligent decision support system, and an agricultural data visualization analysis platform. On this basis, a risk management and operational optimization scheme tailored for the agricultural supply chain will be formed. Through continuous collection and analysis of environmental, soil, growth, and management data during agricultural production, the system can achieve real-time monitoring and dynamic early warnings for key production links, thereby providing more scientific management bases for agricultural enterprises.

From a practical application perspective, this study helps optimize irrigation management, water-fertilizer inputs, and pest and disease control strategies, lowering resource waste and operational risks during agricultural production, and enhancing agricultural product quality stability and market competitiveness. Meanwhile, the data-driven decision support mechanism can also strengthen agricultural enterprises' responsiveness to market demand changes and supply chain risks, elevating overall operational efficiency and asset management levels. From an academic research perspective, this study introduces decision-aware learning, small-sample robust statistical inference, and spatial randomized block experimental design methods into smart agriculture scenarios. It breaks through the limitations of traditional agricultural research that primarily focuses on prediction accuracy, further extending research objectives to economic return optimization and management decision support levels. The analytical framework proposed in this study not only enriches the theoretical system of digital agriculture and agricultural supply chain management but also provides a new research paradigm and practical path for the application of IoT big data technology in agricultural asset management and intelligent decision-making.

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